

Derbyshire Shared Care Pathology Guidelines

Understanding IgE mediated Allergy and requesting Specific IgE blood tests in Adults

Purpose of Guideline

To provide information about the appropriate use of specific IgE (splgE) requests in primary care.

- [Specific IgE requesting by clinical condition](#)
- [In which clinical scenarios should splgE NOT be requested?](#)
- [Interpreting specific splgE and skin prick test results](#)
- [Total IgE requesting](#)

Detailed [background information](#), giving an overview of IgE mediated allergy and how it presents, can be found at the end of the guideline.

IgE mediated allergic conditions include:

- allergic asthma
- allergic rhinitis
- anaphylaxis
- systemic reactions to bee and wasp venom
- some forms of food allergy
- immediate latex allergy
- some forms of drug allergy

Do not request a splgE test unless the clinical history is consistent with possible IgE mediated allergy

Specific IgE requesting by clinical condition

- Asthma/rhinitis: Request splgE to common aeroallergens only if the result will change clinical management. Examples: removing a pet from the home or to meet the criteria for referral for immunotherapy to treat grass pollen, tree pollen or house dust mite induced rhinitis ([Allergic Rhinitis pathway inc Dymista.pdf \(derbyshiremedicinesmanagement.nhs.uk\)](#) and [allergic-rhinitis-primary-care-pathway.pdf \(nottsapc.nhs.uk\)](#))
- House dust mite (HDM) induced asthma or rhinitis: Request splgE to HDM in the context of considering suitability for house dust mite immunotherapy for severe allergic perennial rhinitis ([allergic-rhinitis-primary-care-pathway.pdf \(nottsapc.nhs.uk\)](#) and [Allergic Rhinitis pathway inc Dymista.pdf \(derbyshiremedicinesmanagement.nhs.uk\)](#)) or if patient wishes to consider house dust mite avoidance strategies. Please sign post to Allergy UK guidance and advise them that a pragmatic cost-mindful approach be taken to this [House Dust Mite Allergy | Allergy UK | National Charity](#).

- Bee and wasp venom allergy: Request sIgE to bee and wasp venom only in the context of systemic reactions to a sting and if advised to ahead of an allergy clinic appointment. See section on [bee and wasp venom allergy](#) within background information for definition of a systemic reaction versus large, localised reaction.
- Drugs: DO NOT TEST. SIgE assays for investigating immediate reactions have low sensitivity and are not available to many drugs. Most cases of suspected drug allergy can be managed in primary care with avoidance of the drug and prescription of a safe alternative if required. Readers are advised to review national guidance which outlines a simple clinical classification of allergic drug reactions and their management ([Overview | Drug allergy: diagnosis and management | Guidance | NICE](#)). Where required, please refer to Nottingham University Hospitals Allergy service or Sheffield Teaching Hospitals Allergy service, according to their local referral criteria which can be found on the NHS e-Referral System.
- Latex: DO NOT TEST. Refer to allergy clinic if history consistent with IgE mediated allergy.
- Food allergens. DO NOT TEST. Consider first if the clinical history is suggestive of IgE mediated allergy. See [here](#) for clinical presentations of IgE mediated food allergy and to determine if referral to the allergy clinic is required. If history not suggestive then advise the patient that sIgEs are not indicated.

In which other clinical scenarios should sIgEs NOT be requested?

The following symptoms/scenarios are NOT suggestive of IgE mediated food allergy:

- Eczema in adults
- Fatigue, headache or other non-specific symptoms after foods
- Behavioural/emotional changes
- Bowel symptoms such as bloating, abdominal pain or diarrhoea (outside of the context of an anaphylactic reaction), faecal urgency, weight loss, reflux*
- Day-to-day asthma or rhinitis symptoms outside of the context of an acute anaphylactic/allergic reaction*
- Urticaria (hives and/or angioedema) which occurs at random times, hours after meals and with no clear reliable food culprit and/or occurs on waking (excluding food as a cause). Give consideration when taking a history to the small possibility of wheat-dependent exercise-induced urticaria (see [here](#)), but usually the diagnosis is chronic spontaneous urticaria. Guidance on managing urticaria and angioedema can be found here [urticaria-primary-care-pathway.pdf \(nottsapc.nhs.uk\)](#).

*Patients may have an *intolerance* to certain foods, e.g., spicy foods may cause rhinorrhoea, or wheat may cause bloating in patients with irritable bowel syndrome. SIgE testing has no role here and may cause inappropriate food exclusion. Consider need for investigation for gluten intolerance/inflammatory bowel disease etc in patients with bowel symptoms.

Interpreting sIgE results (and skin prick test results)

The result must always be interpreted in the context of the clinical history.

Key points to understand:

- sIgE tests and skin prick tests are surrogate markers, in the blood and skin respectively, for what is going on in the body or organ involved in a previous reaction. This is one reason that a negative test result does not exclude allergy. Further investigation may be needed and false reassurance may be given if this result alone is relied upon.
- *Sensitisation*, i.e., having a positive test, does not automatically mean the patient will react to that allergen upon exposure (i.e. be *allergic*). This is especially true in atopic people who are often poly-sensitised. There are several possible reasons for this. 1) They have IgE antibodies but another part of their immune system which is not being measured with this test is maintaining tolerance 2) the allergen in the test tube cross reacts with an allergen to which they are allergic. For example, grass pollen allergic patients may have a positive wheat specific IgE but eat wheat with no problem. Their IgE against grass is simply binding to the wheat allergen within the test tube. Positive test results may lead to inappropriate food avoidance and patient anxiety.
- The grade of the sIgE result does not correlate with the severity of any reaction they might have upon exposure. The only useful information given by grade is that patients with high levels of IgE to an allergen are more likely to be allergic to that allergen rather than just sensitised.

Grade	Units (Ku/l)	Interpretation
0	<0.35	Negative
1	0.35 - 0.7	Weak positive
2	0.7 - 3.5	Positive
3	3.5 - 17.5	Positive
4	17.5 - 50.0	Strong positive
5	50.0 - 100	Strong Positive
6	>100	Strong Positive

Total IgE requesting

A total IgE assay measures the total concentration of all IgE molecules circulating in the blood. It is rarely of clinical value and does not need to be requested in a primary care setting. A total IgE within the "normal" range does not exclude IgE mediated allergy. A raised total IgE does not confirm that a reported reaction is IgE mediated in nature. Grossly raised total IgE (often over 1000 kU/L) may be seen in the context of uncontrolled eczema or allergic bronchopulmonary aspergillosis.

Background information

What is IgE mediated allergy?

In susceptible individuals the immune system may begin to produce IgE antibodies to a potential allergen. Allergens are usually proteins and can be found in inhaled substances (aeroallergens), foods, drugs, animal dander, natural rubber latex or insect venom (usually bee or wasp). Initial production of IgE antibodies, known as *sensitisation*, relies on previous exposure to the allergen. Usually the patient has inhaled, ingested or been administered the allergen previously, although sensitisation can also occur via the skin or through breast milk. On subsequent exposure those IgE antibodies recognise the allergen and bind to it. Allergen/IgE binding leads to the rapid release of chemical mediators from allergy immune cells, such as mast cells, found in the mucosa of the body and the skin, and basophils. These chemical mediators, which include histamine, tryptase and leukotrienes, lead to various clinical problems as described below.

Who is susceptible to developing IgE mediated allergy?

- *Atopy* is the genetic predisposition to develop one or more IgE mediated conditions, such as childhood eczema, allergic asthma, allergic rhinitis, immediate food allergies and animal dander allergy. Family history is usually positive.
- *Repeated allergen exposure*. Patients most likely to develop IgE mediated drug allergic reactions are those who repeatedly receive the same drug, for example patients with cystic fibrosis requiring regular penicillins for exacerbations of their lung disease. Anaphylaxis to bee venom is uncommon outside of the context of beekeepers, who may become sensitised after being stung repeatedly. These patients do not have to be atopic.

What are the hallmarks of an IgE mediated reaction?

- When the reaction occurs after a one-off exposure, for example with foods, drugs or close contact with animal dander, symptoms generally begin **immediately or within the first hour, and are rapidly evolving**. Immediate/IgE mediated drug allergy may present within four hours. Important caveats to this hallmark are highlighted [here \(delayed co-factor food induced anaphylaxis, page 7\)](#)
- The reaction usually **occurs each time the patient is exposed to the allergen** (unless a co-factor is required as outlined [here \(delayed co-factor food induced anaphylaxis, page 7\)](#)).
- In allergic asthmatic or allergic rhinitis patients with daily exposure it may be harder to determine which common aeroallergens are driving symptoms. Allergy testing may be indicated if the result will change management.

How can IgE mediated allergy present clinically?

1. Asthma, seasonal allergic rhinitis or perennial allergic rhinitis

Tree, grass, weed pollens and some mould spores can drive seasonal allergic rhinitis. Seasonal timing can often identify specific allergen.

Trees – March-April

Grass – May-September

Weeds – autumn

Moulds – perennial or autumn symptoms

Direct contact with grass pollen or animal saliva may also cause contact urticaria.

Perennial (year-round) allergic rhinitis is commonly caused by allergy to house dust mite faeces or animal dander allergy. Symptoms from house dust mite allergy are often worse in the night and first thing in the morning and tend to improve in hot or cold climates where the mite cannot survive. Symptoms may be induced by allergens found in animal dander and saliva, commonly from cats, dogs, rabbits, horses and guinea pigs. Proteins in cat dander are highly allergenic and spread all over the house on clothing. Affected individuals are likely to be allergic to all cats, including their own, even if they feel they have developed a level of tolerance. Allergy to dogs is more variable by person and by dog breed, but no dog can be truly said to be "hypoallergenic" as allergens are found in saliva as well as dander.

In adults, both asthma and perennial rhinitis may be non-allergic in origin. Inflammation, usually eosinophilic, is still present and in most patients the treatment is similar as for those with allergic disease.

- Allergy testing is generally only relevant in those patients at the more treatment resistant end of the clinical spectrum, who might benefit from targeted treatments. In the case of asthma this might be omalizumab (anti-IgE monoclonal antibody) to reduce exacerbations. In severe allergic rhinitis, a 3-year course of desensitisation to grass pollen, tree pollen or house-dust mite, using sublingual or subcutaneous immunotherapy may be indicated if the patient is mono-sensitised and reacting to one allergen. Local guidelines for managing rhinitis can be found here ([Allergic Rhinitis pathway inc Dymista.pdf](#) ([derbyshiremedicinesmanagement.nhs.uk](#)) and here [allergic-rhinitis-primary-care-pathway.pdf](#) ([nottsapc.nhs.uk](#)))

2. Anaphylaxis

This is the most severe presentation of an IgE mediated allergy. The commonest causes are due to allergens within foods (e.g. nuts, shellfish, milk, wheat), drugs or bee or wasp stings. Rarely, it can arise spontaneously or by non-IgE mediated mechanisms.

Resuscitation Council UK has published guidelines on the diagnosis and immediate management of anaphylaxis ([Emergency treatment of anaphylactic reactions: Guidelines for healthcare providers | Resuscitation Council UK](#)).

All cases should be referred to an allergy clinic for further investigation and management, in line with NICE recommendations for management of anaphylaxis. Please refer in with as much clinical detail of the reaction as possible, ideally with a copy of the emergency department notes, and with any available serum tryptase results, with timings

([Overview | Anaphylaxis: assessment and referral after emergency treatment | Guidance | NICE](#)).

See here for further information on prescribing adrenaline auto-injectors (AAIs) in primary care:

[Adrenaline auto-injector prescription for patients at risk of anaphylaxis: BSACI guidance for primary care - Angier - 2023 - Clinical & Experimental Allergy - Wiley Online Library](#)

3. Bee and wasp venom allergy

Insect stings and bites (horseflies) may cause large, localised reactions. These are defined as an area of induration with a diameter of >10 cm and which peaks between 24 and 48 hours and then subsides. Large, localised reactions do not require further investigation or the prescription of an AAI.

Systemic reactions may range from immediate urticaria (hives and/or angioedema) or flushing/itching, through to life threatening anaphylaxis presenting with cardiovascular collapse. All patients with a systemic reaction should be referred to the allergy clinic for testing and diagnosis, training in the use of an AAI and consideration of subcutaneous venom immunotherapy.

In the event of a systemic reaction please prescribe two AAIs to be carried at all times, whilst awaiting allergy clinic review. The allergy clinic may advise testing for IgE to bee and wasp venom.

4. IgE mediated food allergy

There are three main clinical presentations of IgE mediated food allergy to consider:

1. Pollen-food syndrome (previously known as oral allergy syndrome).

Patients with seasonal allergic rhinitis, most commonly to birch pollen in the spring, experience immediate allergic symptoms when eating some fruits, vegetables or nuts. Symptoms include oral itching, minor tongue or lip swelling and occasionally throat swelling. Culprit foods are tolerated in their peeled or cooked forms. In this syndrome the patient has IgE antibodies to a pollen allergen which is structurally like allergens found within these foods. On ingestion the IgE antibodies recognising pollen cross react with those food allergens, leading to mast cell degranulation. The cross-reacting allergens are often found in the peel only, or broken down by gut enzymes or cooking, hence the patient can tolerate the food when peeled or cooked. SptIgE testing for pollens is helpful but is not required for the actual foods, in the presence of a classical history. The patient should be advised to avoid the culprit fruit or vegetables in their raw form and to avoid smoothies containing those foods (as large amounts of concentrated allergen may cause a systemic reaction). If nuts are involved these should be avoided in all forms and the patient should be referred to an allergy clinic to determine if their nut allergy is exclusively to pollen food syndrome or due to primary nut sensitisation. The British Society of Allergy and Clinical Immunology (BSACI) have published further guidance for primary care here https://www.bsaci.org/wp-content/uploads/2022/08/Pollen-Food-Syndrome-leaflet-for-GPs_Aug2022.pdf

2. Primary food allergen sensitisation.

The patient has developed IgE antibodies to a food allergen rather than pre-existing IgE antibodies to pollen allergens cross-reacting to structurally similar food proteins. The food allergens involved are heat-stable and able to make their way to the gut and hence are more likely to cause systemic and potentially severe allergic reactions. Common examples of foods which induce reactions include peanuts and tree nuts, milk (children generally), shellfish, sesame and wheat, but many more are

implicated. In general symptoms develop within minutes to an hour of exposure and may include any of the following:

- Urticaria (hives and/or angioedema).
- Flushing and itch.
- Throat swelling, stridor, voice change (laryngeal angioedema).
- Feeling faint, suggesting hypotension.
- Wheeze and chest tightness
- Abdominal cramps, nausea, vomiting and diarrhoea.
- Sense of impending doom

The patient should be prescribed two AAls to be carried at all times, whilst awaiting allergy clinic review. Allergy testing, sptIgEs and/or skin prick testing can be performed in the allergy clinic.

3. Co-factor (sometimes delayed) food-induced anaphylaxis.

In this uncommon syndrome patients have IgE antibodies to a food allergen but only develop allergic symptoms when another co-factor is present. Common co-factors include exercise, use of non-steroidal anti-inflammatory drugs, intercurrent illness, or alcohol ingestion. The reaction may also be delayed by up to 4 hours after the ingestion of the food. Commonly the IgE antibodies are against a specific allergen within wheat (omega 5 gliadin, which causes wheat-dependent exercise induced anaphylaxis) or against a group of food allergens found in plant foods, known as lipid-transfer proteins (LTPs). Clinicians should consider this syndrome in patients where anaphylaxis appears to be occurring spontaneously and patients should be referred to an allergy clinic for investigation. AAls should be prescribed.

5. Natural rubber latex allergy

Latex allergy is less common now with the phasing out of latex gloves in most healthcare settings. Can be associated with:

1. Immediate IgE mediated symptoms- mild hives through to anaphylaxis. Refer to allergy clinic if suspected.
2. Food-latex cross reactions. Similar to pollen food syndromes, the IgE antibodies in patients with latex allergy may cross react with proteins in certain foods, e.g., banana, chestnut, avocado. Refer to allergy clinic if suspected.
3. Allergic asthma if exposed regularly to powdered latex gloves.
4. Delayed contact dermatitis at site of exposure (investigated by dermatology)
5. Irritant contact dermatitis at site of exposure (not due to delayed cell mediated allergy).

Contacts

For clinical advice: please use Sheffield, Nottingham or Birmingham advice (Depending on patient location) and guidance on e-RS.

For laboratory advice / result interpretation: Dr Aarnoud Huissoon (Consultant Immunologist at Derby Immunology Laboratory and Birmingham Heartlands Hospital).

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References

Emergency treatment of anaphylaxis, Guidelines for healthcare providers. Resuscitation Council (UK), May 2021 [Emergency treatment of anaphylactic reactions: Guidelines for healthcare providers | Resuscitation Council UK](#)

Anaphylaxis: assessment and referral NICE guidance [Overview | Anaphylaxis: assessment and referral after emergency treatment | Guidance | NICE](#)

[Adrenaline auto-injector prescription for patients at risk of anaphylaxis: BSACI guidance for primary care - Angier - 2023 - Clinical & Experimental Allergy - Wiley Online Library](#)

Drug allergy: diagnosis and management NICE guideline [Overview | Drug allergy: diagnosis and management | Guidance | NICE](#)

Allergy UK, House dust mite fact sheet [House Dust Mite Allergy | Allergy UK | National Charity](#)

UK Allergy Guidelines for primary care from BSACI (including venom allergy, urticaria, rhinitis, drug allergy) [Primary Care Guidelines - BSACI](#)

https://www.bsaci.org/wp-content/uploads/2022/08/Pollen-Food-Syndrome-leaflet-for-GPs_Aug2022.pdf

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